Problem 1: Strange WRIGGLE Reports

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2025-11-10

Setup

Your research team has arrived at the Worldwide Observatory for Research on Macroparasite Stuff (WORMS), where you're excited to begin your summer research internship. Before you've even had a chance to sit down at your new desk and report to your first-day orientation, you're immediately intercepted by a group of anxious technicians from the Nematode Observation Division.

They explain that overnight their Worm Information Gathering & Global Logging Engine (WIGGLE) has started producing unusual data streams originating from central South Carolina. Since you're already here (and apparently have a reputation for debugging strange biological systems), they ask for your help making sense of it.

You glance toward the rest of your research group, but they've already dispersed—half of them already getting started centrifuging egg samples, the other half arguing over the calibration of a worm treadmill. You sigh and agree to take a look.

The Data

The unusual data (your puzzle input) consists of multiple "nematode activity" reports, one per line. Each report represents a short recording of "nematode activity levels" (an ambiguous term, but the technicians insist it's useful) at a given site, with each recording separated by a space.

For example:
7 6 4 2 1
1 2 7 8 9
9 7 6 2 1
1 3 2 4 5
8 6 4 4 1
1 3 6 7 9

The Question

The technicians want to identify which reports indicate stable soil nematode populations.

A report is considered stable if both of the following conditions are true:

1. The nematode activity levels are consistently increasing or consistently decreasing (no reversals).
2. Any two adjacent levels differ by at least 1 and at most 3 units.

These constraints reflect normal fluctuations of nematode populations in the environment. Larger jumps suggest disturbance-associated shifts or even a potential helminth-driven disease outbreaks.

Example Interpretation
In the example above:
* The first report (7 6 4 2 1) shows a smooth decrease — stable.

* The second (1 2 7 8 9) contains a sudden jump of 5 — unstable.
* The third (9 7 6 2 1) unstable because 6 2 is a decrease of 4.
* The fourth (1 3 2 4 5) unstable because 1 3 is increasing but 3 2 is decreasing.
* The fifth (8 6 4 4 1) unstable because 4 4 is neither an increase or a decrease.
* The final (1 3 6 7 9) stable because the levels are all increasing by 1, 2, or 3.

So, in this example, 2 reports are stable.

The data from all sensors has been collected by the technicians and compiled into the file "input1.txt"

How many of these reports are safe?

Part II:

Later, a postdoc points out that some fluctuations might be due to sensor noise or interference. Maybe the nematode activity pattern could still be considered stable if one data point per report is ignored.

How many reports total reports are considered safe if you assume one reading can be discarded from an unsafe reports?